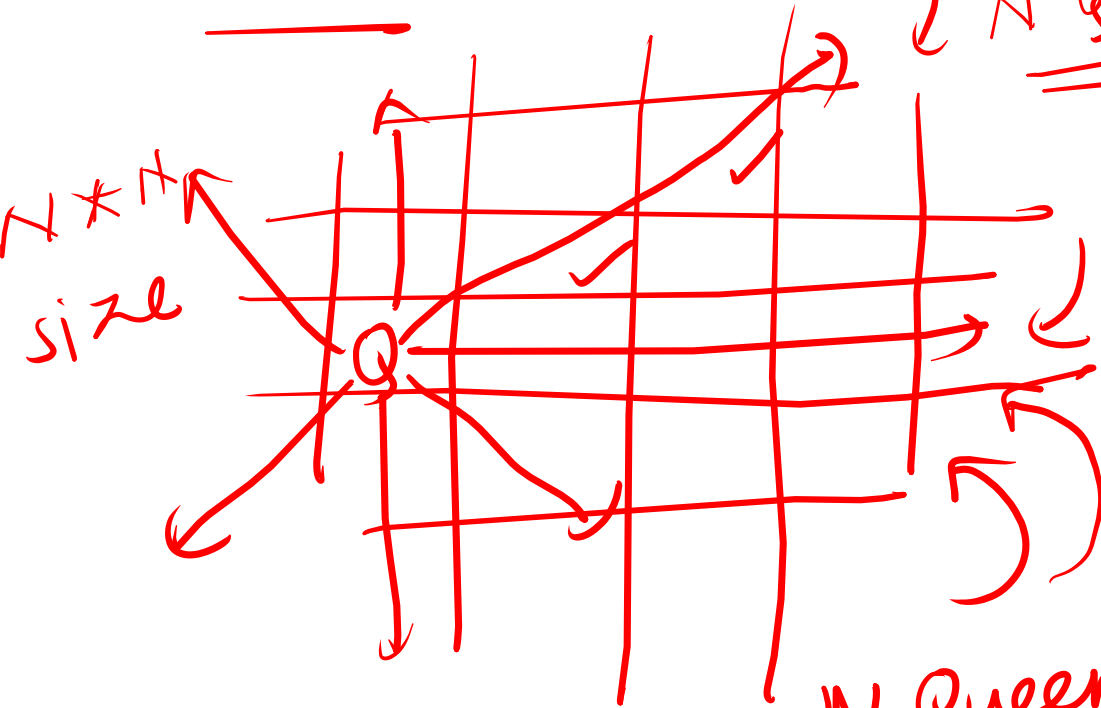


N Queen ✓✓

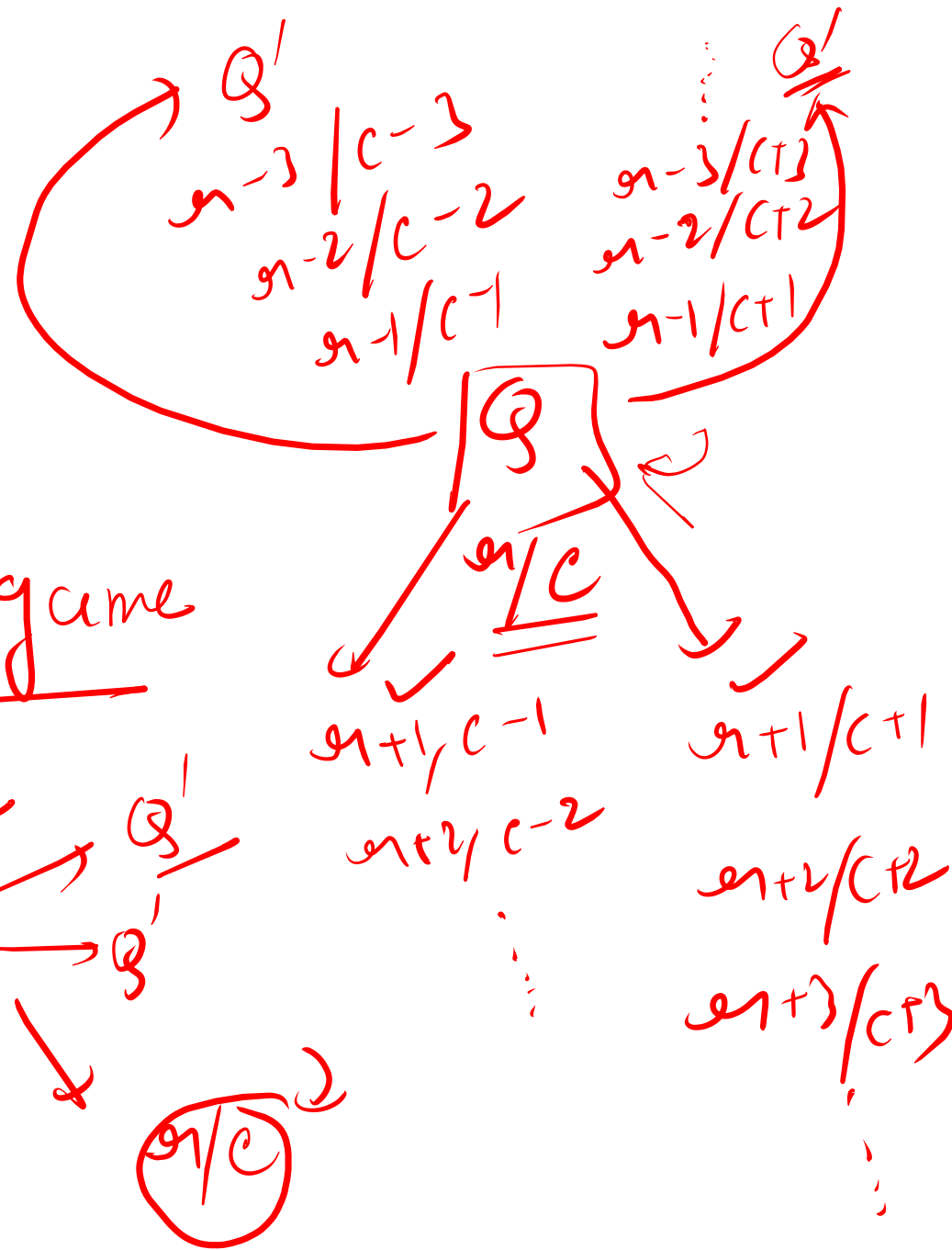
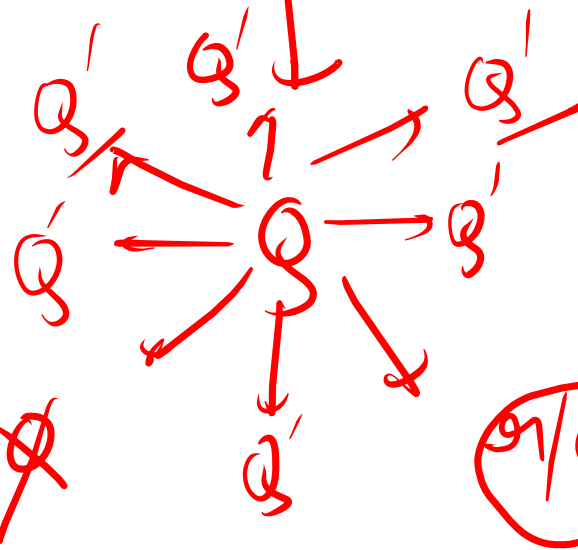
N = 4

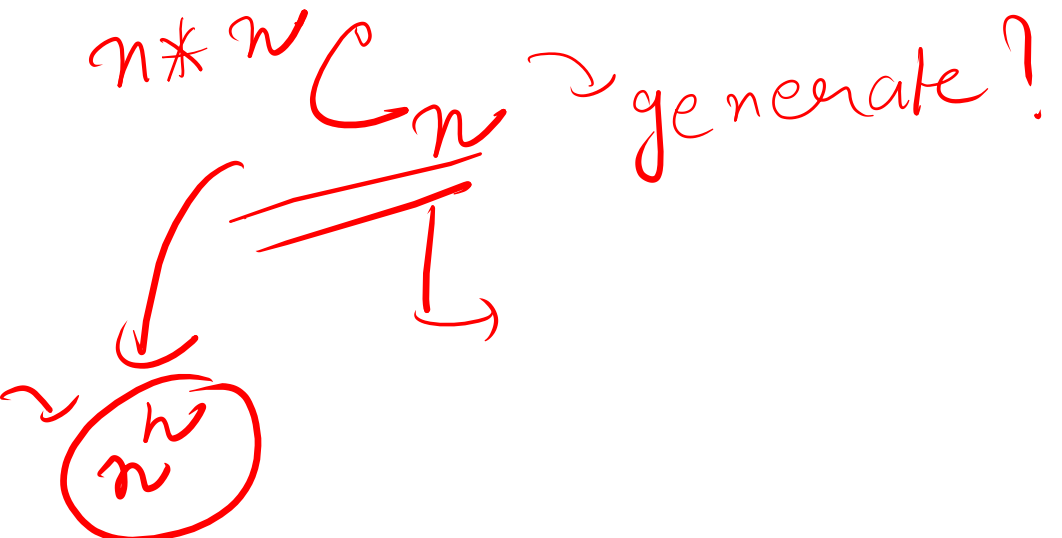
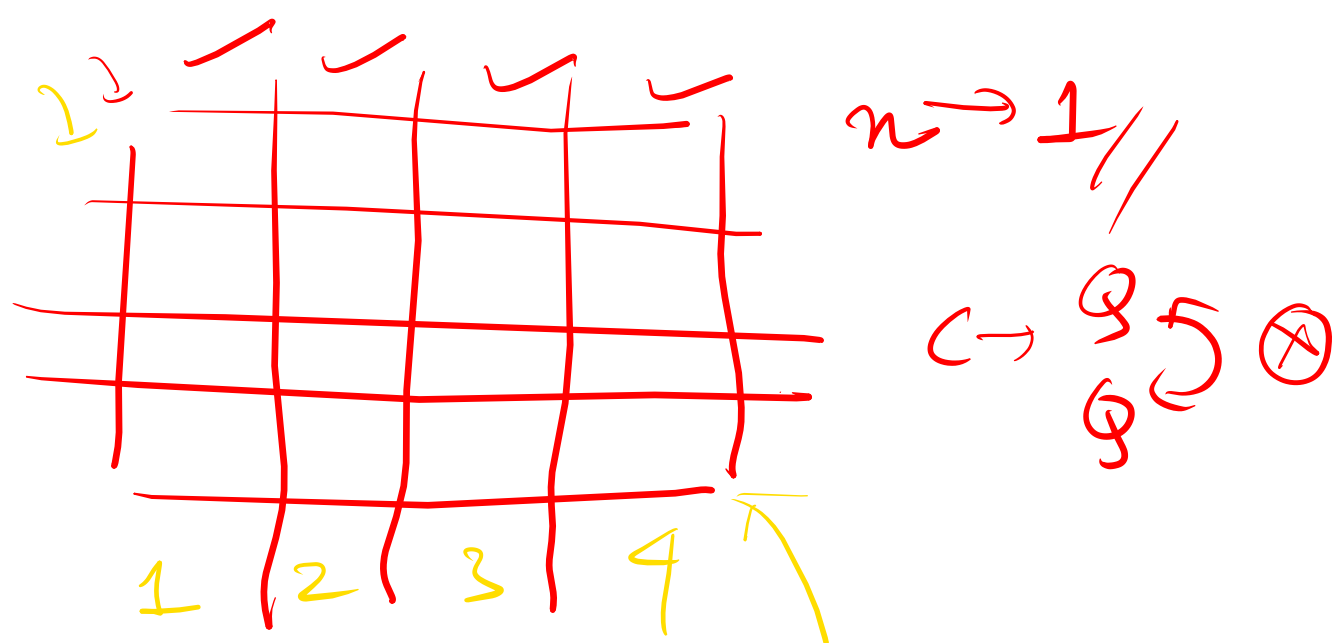
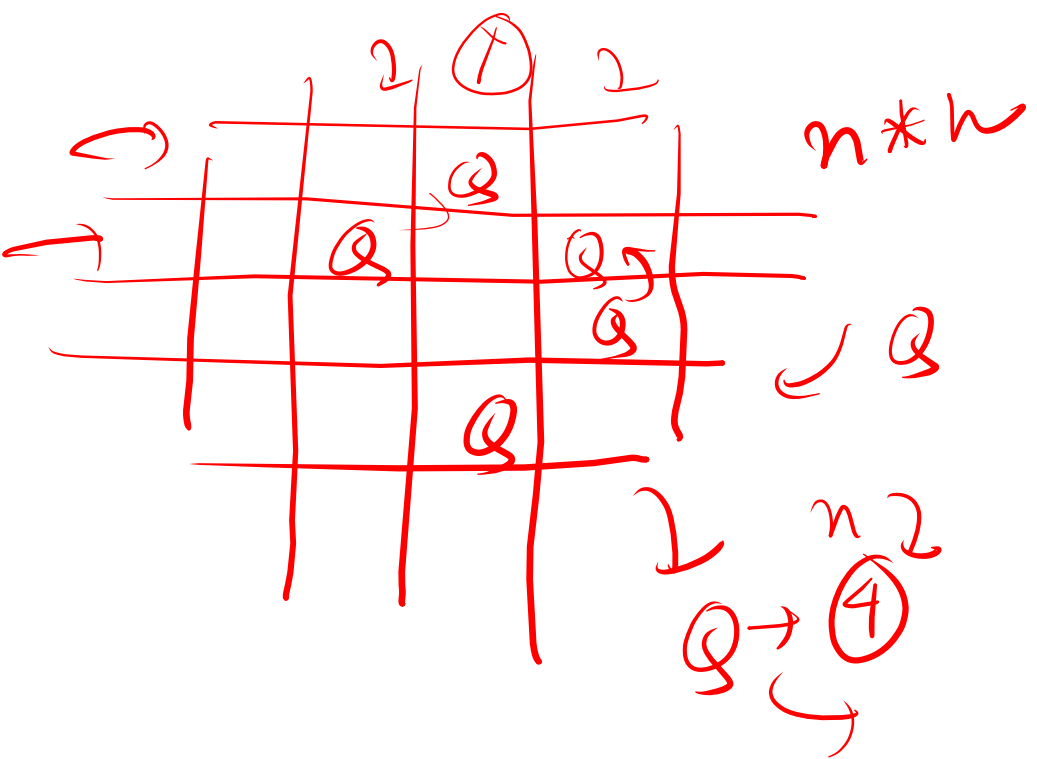


N Queens

N Queens
No 2 Queens
→ attack

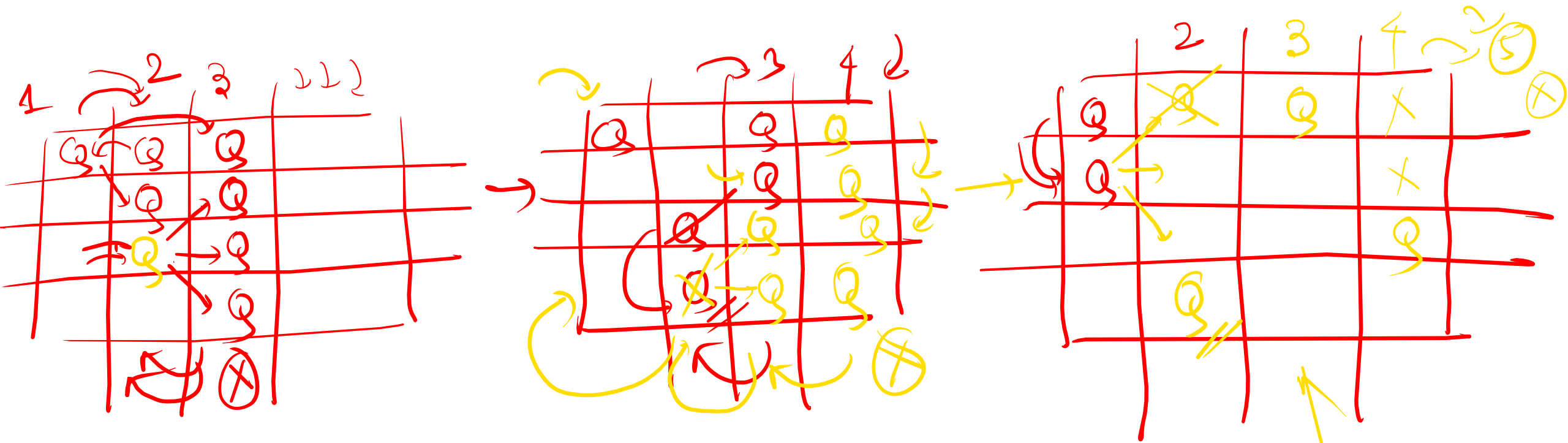
chess game





1 2

Q

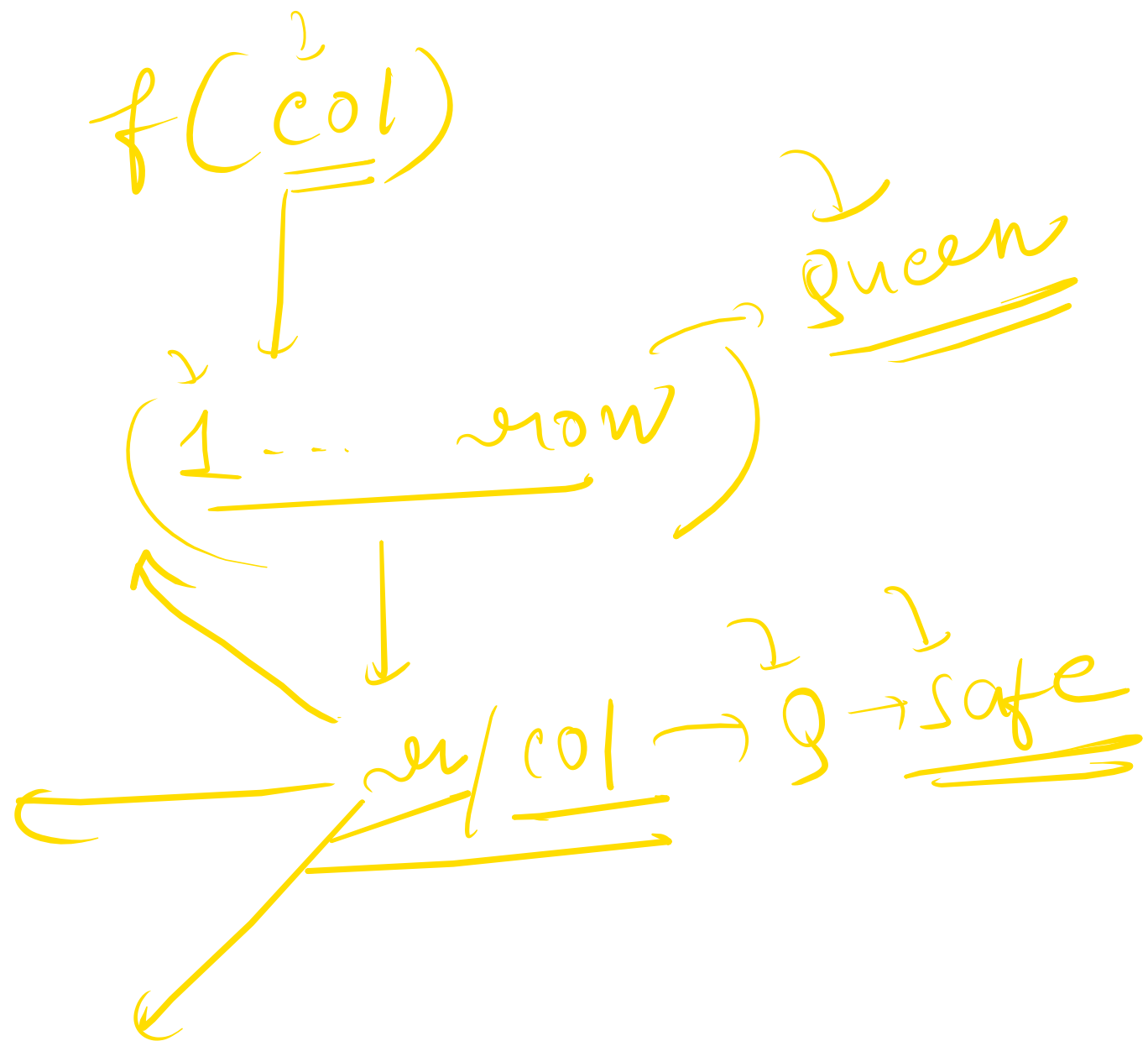
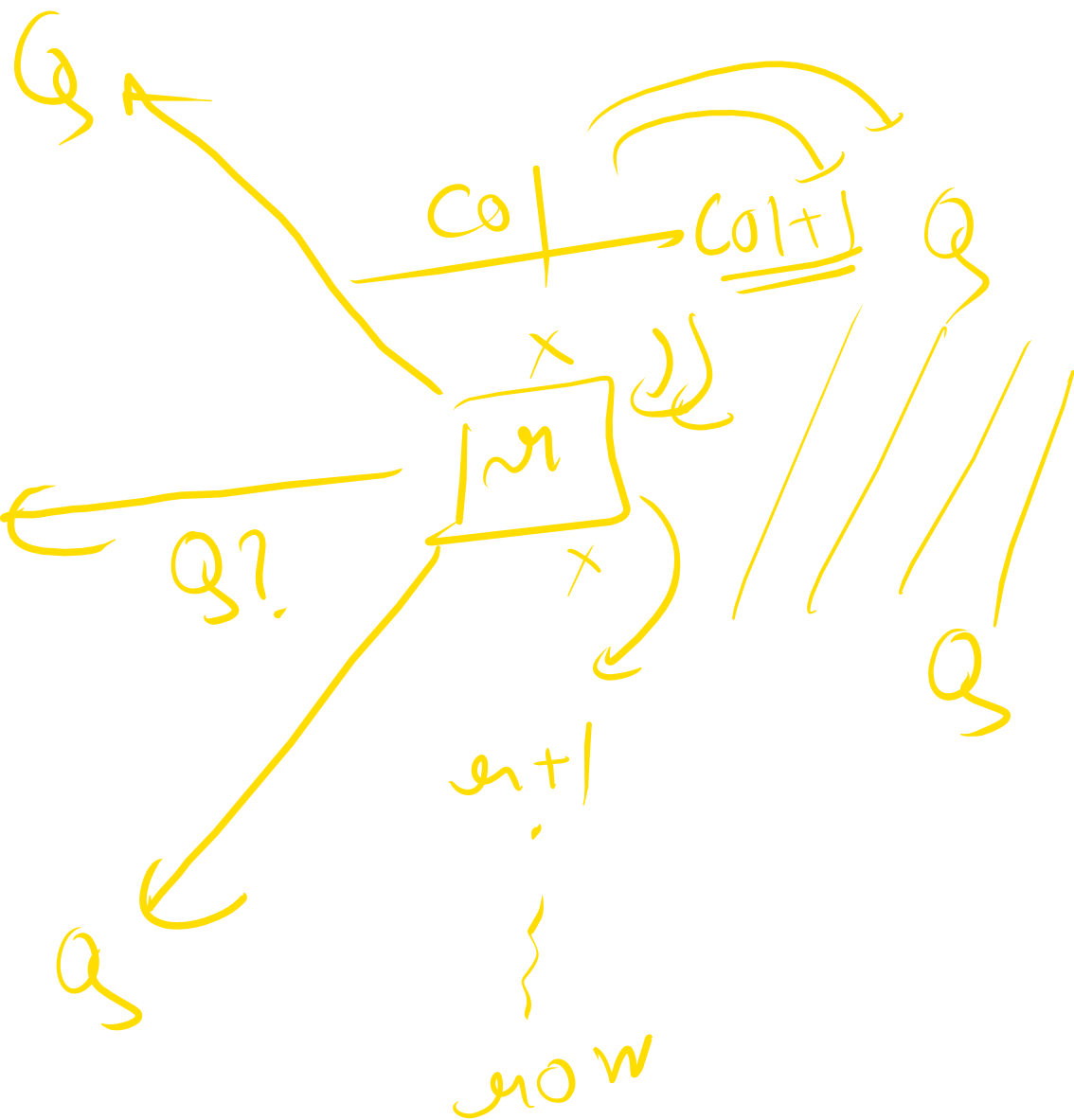


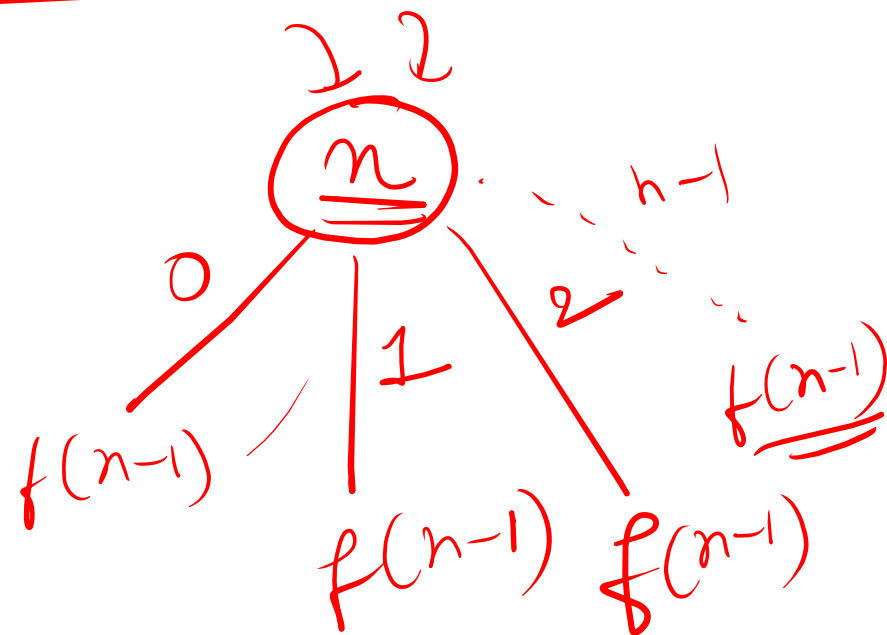
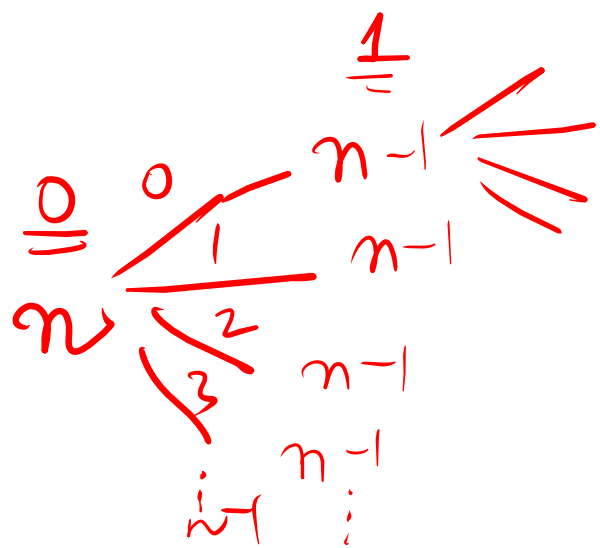
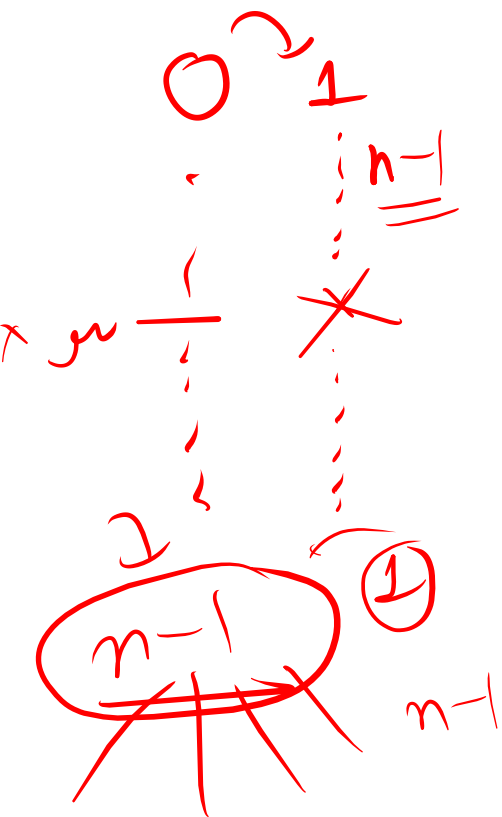
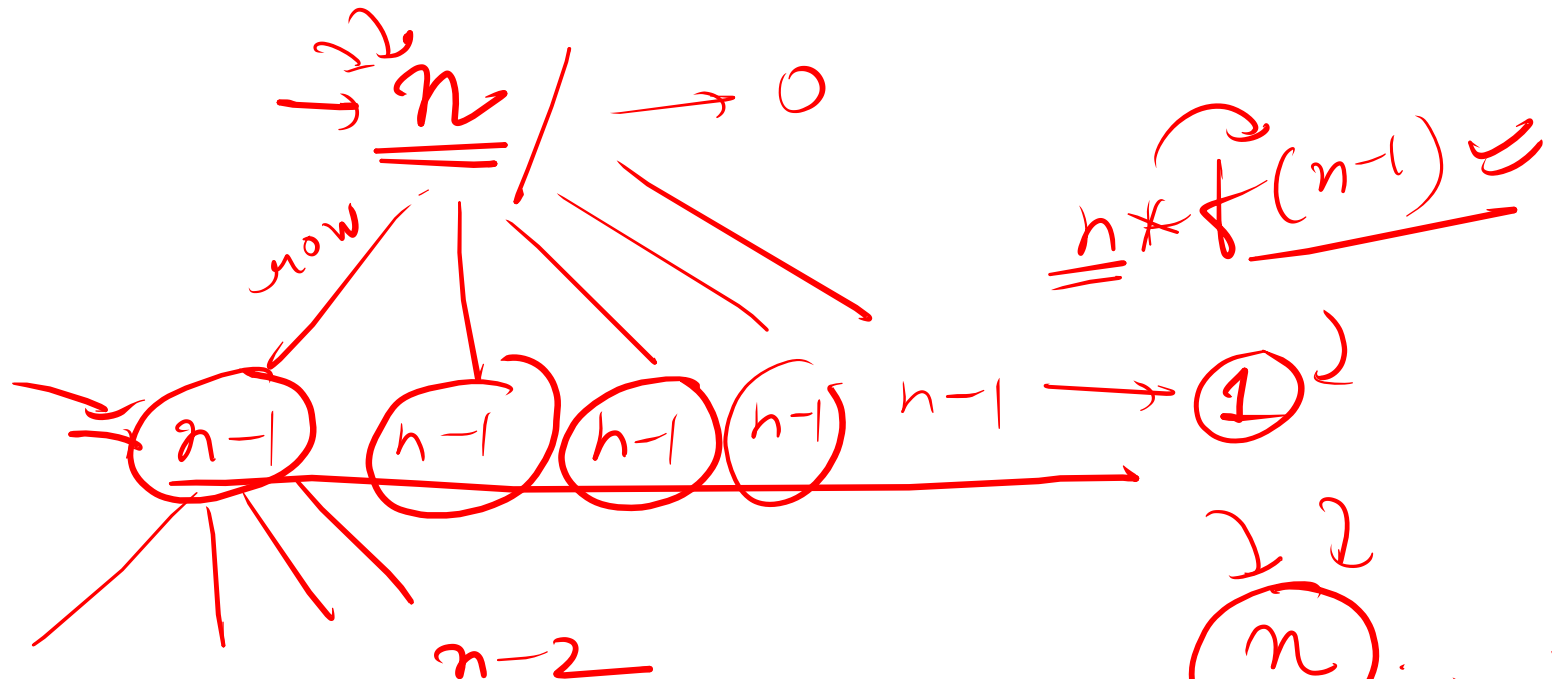
1/1 $\rightarrow$ Q $\rightarrow$ how many

soln $\rightarrow$ Optimized

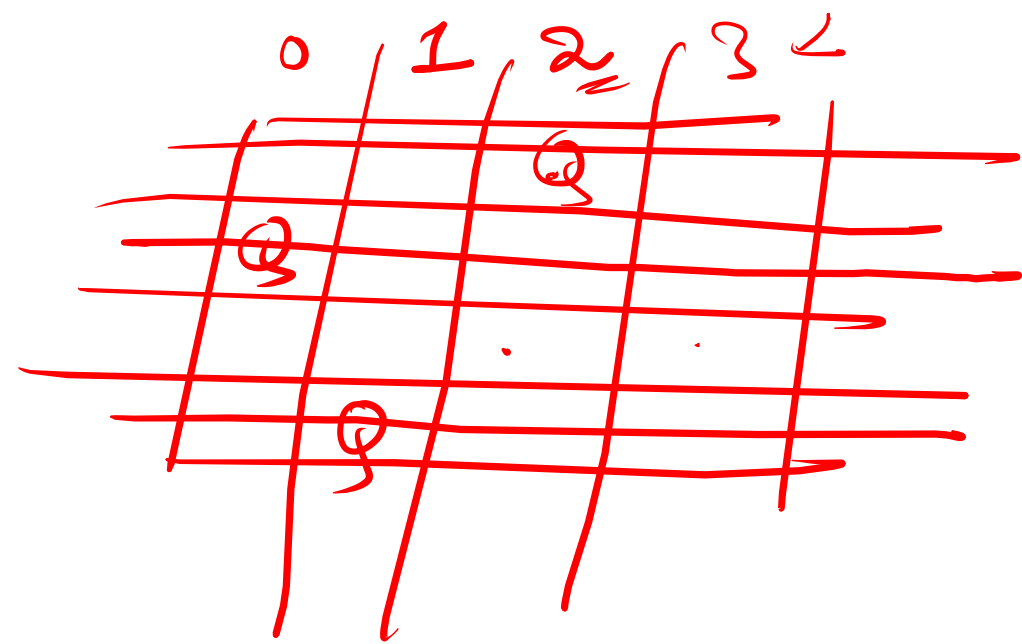
~~1/2~~ 1/2 $\rightarrow$ Q $\rightarrow$ Backtracking $\rightarrow$ Q or/c

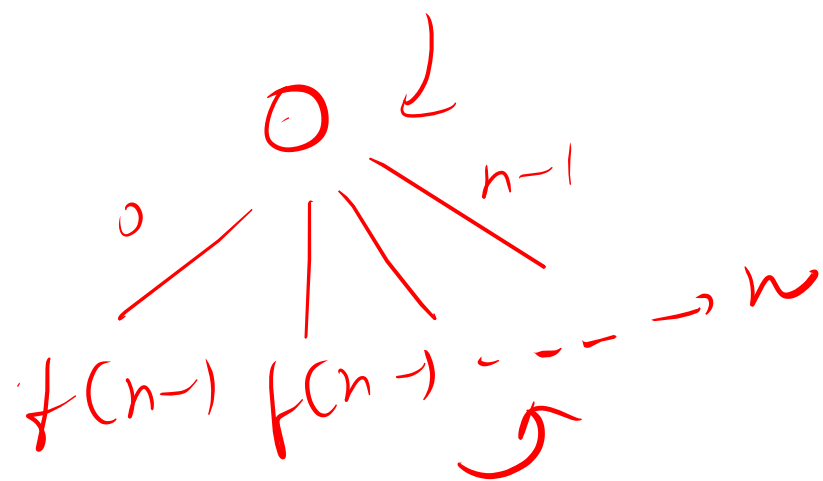
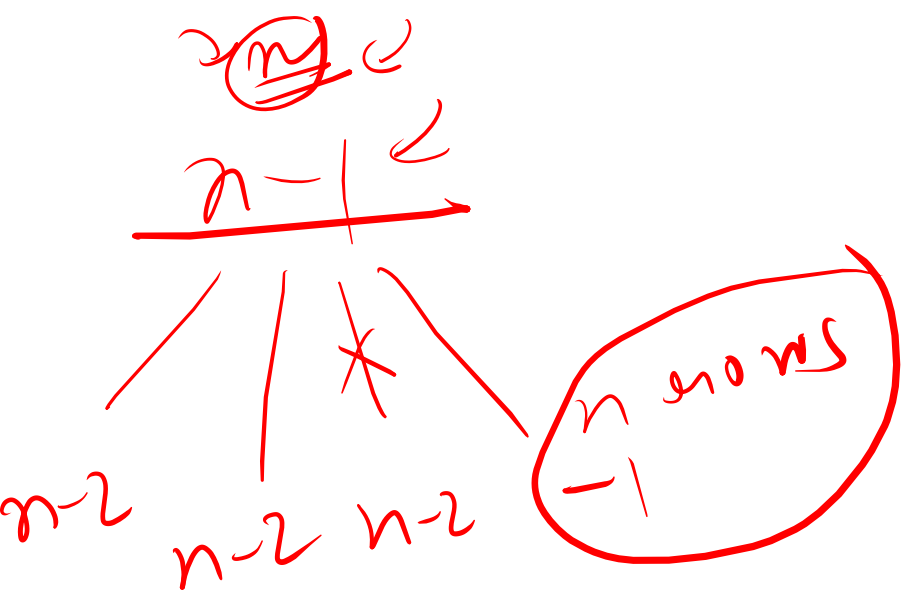
solution





$\checkmark$ $n \rightarrow 0 \rightarrow n \text{ rows}$
 $\checkmark$ $n-1 \rightarrow 1 \rightarrow n-1 \text{ rows}$
 $n-2 \rightarrow 2 \rightarrow n-2 \text{ rows}$
 $\vdots$
 $1 \rightarrow n-1 \rightarrow 1 \text{ row}$





$$T(n) = n * T(n-1)$$

$$T(n-1) = (n-1) * T(n-2)$$

$$T(n-2) = (n-2) * T(n-3)$$

$$T(n-3) = (n-3) * T(n-4)$$

⋮

$$T(1) \rightarrow 1 * K$$

$$T(n) = \underline{n} * \underline{n-1} * T(n-2)$$

$$\downarrow$$

$$\underline{n-2} * T(n-3)$$

$$\geq n * n-1 * n-2$$

⋮

$$* 1$$

$$\downarrow$$

$$\underline{\underline{O(n!) \rightarrow \underline{\underline{IC}}}}$$

$$\downarrow$$

$$\underline{n-3} * T(n-4)$$

⋮

$$\underline{1} * \underline{T(0)}$$

$$\downarrow$$

$$\textcircled{K}$$

Handwritten diagram illustrating the steps of the Euclidean algorithm for $\gcd(21, 13)$:

- $21/13$
- $13/6$
- $6/5$
- $5/1$
- $1/0$

The diagram shows the sequence of divisions and the resulting remainders, leading to the final result of 1.

$\% \quad \begin{array}{cccc} & 0 & 1 & 2 & \textcircled{3} \\ \hline & 1 & 3 & 0 & \end{array}$

$$(\begin{bmatrix} \overset{22}{\underset{0}{\equiv}} & \underline{1} & \underline{2} \\ 1 & \underline{3} & 0 \end{bmatrix}^2$$

er

$(er, 3)$

3

er

$$\left| \frac{x - (x+2)}{2} \right| / \left| \frac{c - (c-2)}{2} \right|$$

A handwritten diagram in red ink illustrating the distance between two points. At the top, two points are labeled (x, y) and (x', y') . A horizontal line segment connects them, with a double-headed arrow below it. Below this, the expression $|x - x'| = |y - y'|$ is written, with a double-headed arrow pointing to the right-hand side of the equation.



for array
= c-i

not
safe

true